RESEARCH STATEMENT

SHISHIR AGRAWAL

1. Introduction

1.A. **Panorama.** My research studies interactions between algebraic geometry in positive characteristic p and p-adic analytic geometry. This branch of mathematics began to blossom following Dwork's seminal application of p-adic methods to the first of the Weil conjectures, the rationality of the zeta function of a variety over a finite field [Dwo60]. Monsky and Washnitzer constructed a rational p-adic cohomology theory to "explain" Dwork's proof [MW68]. Its modern incarnation is Berthelot's rigid cohomology [Ber83].

These ideas have taken root across a wide swath of mathematics. In coding theory and cryptography, they underlie efficient algorithms for counting points on varieties over finite fields, as in Kedlaya's algorithm for hyperelliptic curves [Ked01] and its generalization by Lauder and Wan [LW08]. In pure mathematics, some of the newest developments tie into representation theory, as in Abe's p-adic Langlands correspondence [Abe13] and the Beilinson-Bernstein style localization theory due to Huyghe, Patel, Schmidt and Strauch [HPSS15].

The most mature theory of a category of coefficients for rigid cohomology is that of arithmetic D-modules, objects which roughly correspond to systems of p-adic differential equations. The groundwork for the theory was laid by Berthelot [Ber96], and its development has been the content of a vibrant research program over the past two decades. The role of "smooth" objects among arithmetic D-modules is played by objects known as overconvergent[1] isocrystals; this is analogous to the role local systems play among constructible sheaves, or that integrable connections play among holonomic D-modules.

My research centers around investigating *rigid* isocrystals on curves, which, roughly, are systems of p-adic differential equations that are determined by their behavior near singularities. More specifically, my work thus far addresses the following:

- Understanding how rigidity relates to infinitesimal deformations of isocrystals.
- Classifying rigid isocrystals in certain situations.

Some of these results can be found in my preprint [Agr17], and others will appear in forthcoming work. This exploration is motivated by analogies with more well-studied notions of rigidity for local systems and integrable connections.

[1]For readability, I will sometimes omit the word "overconvergent." It is always implied.

1.B. **Rigidity for "smooth" objects à la Katz.** Throughout, let X be a smooth proper and geometrically connected curve over a perfect field k, and $j : U \hookrightarrow X$ the inclusion of a dense affine open subset with complement $S := X \smallsetminus U$.

When $k = \mathbb{C}$, Katz defines two versions of rigidity for a local system E on U^{an}. It is *physically rigid* if it is determined up to isomorphism by its local monodromies along S, and *cohomologically rigid* if $H^1(X, j_* \operatorname{End}^0(E)) = 0$, where $\operatorname{End}^0(E)$ denotes the local system of traceless endomorphisms of E [Kat96]. For irreducible local systems when $X = \mathbb{P}^1$, Katz proves that these notions coincide, and that these local systems have very rich structure: they "come from geometry," and they can be "built up" out of rank 1 objects by a procedure called the middle convolution algorithm.

Katz also defines rigidity for ℓ-adic local systems for arbitrary k, where ℓ is a prime distinct from $\operatorname{char}(k)$, and many of his results are proved in this context. When $\operatorname{char}(k) = 0$, Bloch, Esnault [BE04] and Arinkin [Ari08, Ari10] develop a comparable theory of rigid integrable connections. The analogous notion of rigidity for isocrystals when $\operatorname{char}(k) > 0$ was introduced recently by Crew [Cre17].

1.C. **Rapid introduction to overconvergent isocrystals.** Suppose $\operatorname{char}(k) > 0$ and $K := \operatorname{Frac} W(k)$. Let $\mathfrak{X}$ be a smooth formal lift of X over $W(k)$ and $\mathfrak{X}_K$ the associated analytic space. The preimage of $Y \subseteq X$ under the specialization map $\mathfrak{X}_K \to \mathfrak{X}$ is its *tube* $]Y[$. The tube $]x[$ of a closed $x \in X$ is an open disk in $\mathfrak{X}_K$. Thus $]S[$ is a finite union of open disks and $]U[$ is its complement. A neighborhood of $]U[$ is *strict* if it "extends into" $]S[$.

For example, suppose $X = \mathbb{P}^1$ and $S = \{0, \infty\}$. As a set, the analytic space $\mathfrak{X}_K$ (à la Tate) consists of the Galois orbits in $\bar{K}$ plus an additional point ∞. We have the following tubes.

$$]0[= \{x \in \bar{K} : |x| < 1\} \qquad]\infty[= \{x \in \bar{K} : |x| > 1\} \cup \{\infty\} \qquad]U[= \{x \in \bar{K} : |x| = 1\}$$

The annuli $\{x \in \bar{K} : \lambda \leq |x| \leq \lambda^{-1}\}$, where $0 < \lambda < 1$, form a base for the filter of strict neighborhoods of $]U[$.

An *overconvergent function* on U is one that defined on a strict neighborhood of $]U[$. The ring A of all overconvergent functions on U is a topological Dedekind domain; when $X = \mathbb{P}^1$, it is even principal. The A-module $\widehat{\Omega}^1_{A/K}$ of continuous differentials is projective of rank 1.

For any closed $x \in X$, let A_x be the ring of analytic functions on $]x[\cap V$, where V varies over strict neighborhoods of $]U[$. (When $s \in S$, the ring A_s is the *Robba ring* at s.) It is a topological Bézout domain, and there is a continuous "restriction" map $A \to A_x$ which is "étale" in the sense that it is flat and $\widehat{\Omega}^1_{A_x/K} = A_x \otimes_A \widehat{\Omega}^1_{A/K}$. A finite A-module with connection E thus induces a finite A_x-module with connection $E_x := A_x \otimes_A E$ which captures the "local behavior" of E around x.

An *(overconvergent) isocrystal* on U is a finite A-module with connection E whose Taylor series converges in a strict neighborhood of $]\Delta(U)[$ inside $\mathfrak{X}_K \times \mathfrak{X}_K$, where $\Delta : X \to X \times X$ is the diagonal embedding. This condition implies that E_u is constant (i.e., has a full set of

horizontal sections) for any closed $u \in U$. It also implies that E_s is solvable at 1 for any $s \in S$, which *roughly* means that formal horizontal sections of E_s have sufficiently large radii of convergence in disks inside $]s[\cap V$, where V is a strict neighborhood of $]U[$.

We write $\mathrm{Isoc}^\dagger(U)$ for the category of isocrystals on U. Tensor product over A gives it the structure of a K-linear tannakian category. It is independent of the choice of lift $\mathfrak{X}$.

2. Rigid isocrystals

2.A. **Physical rigidity.** Like for local systems, we can define two versions of rigidity for isocrystals. First, $E \in \mathrm{Isoc}^\dagger(U)$ is *physically rigid* if it is isomorphic to any $E' \in \mathrm{Isoc}^\dagger(U)$ such that $E_s \simeq E'_s$ for all $s \in S$.

When $k = \mathbb{C}$, physically rigid local systems on U^{an} can only exist when X has genus 0, because otherwise we can tensor with the restriction of a rank 1 local system on X^{an} of infinite order without changing the local monodromies around S. By a similar argument, I can prove the following.

Proposition 2.1. *Suppose k is algebraically closed*[2] *and that the p-rank of X is nonzero. Then there exists $L \in \mathrm{Isoc}^\dagger(X)$ of rank 1 no tensor power of which is constant. Thus, there cannot exist physically rigid isocrystals on U.*

At the moment, I do not know if this result is true if we only assume that X has nonzero genus (e.g. if X is a supersingular elliptic curve). The construction of the L of proposition 2.1 uses the Katz-Crew equivalence [Cre87, 2.1], but when the p-rank of X is 0, any $L \in \mathrm{Isoc}^\dagger(X)$ of rank 1 obtained through this equivalence will necessarily be of finite order.

2.B. **Cohomological rigidity.** Second, $E \in \mathrm{Isoc}^\dagger(U)$ is said to be *cohomologically rigid* if $H^1_p(U, \mathrm{End}^0(E)) = 0$, where $H^1_p(U, -)$ is Crew's parabolic cohomology: i.e., the image of the "forget supports" map $H^1_c(U, -) \to H^1(U, -)$. I can prove that parabolic cohomology can be reinterpreted using middle extensions of arithmetic D-modules, so this cohomological rigidity condition mirrors the analogous conditions for local systems and for integrable connections.

Proposition 2.2. *When E has a Frobenius structure,*[3] $H^1_p(U, E) = H^1(X, j_{!+}E)$.

Clearly rank 1 isocrystals are always cohomologically rigid. The Christol-Mebkhout index formula [CM00] implies that cohomologically rigid isocrystals of rank at least 2 can only exist when the genus of X is either 0 or 1.

[2]The restriction that k be algebraically closed appears to be significant: if k is finite, any $L \in \mathrm{Isoc}^\dagger(X)$ of rank 1 and infinite order cannot have a Frobenius structure [Abe11, 6.1], so must be a bit pathological.

[3]A comment for non-experts: "having a Frobenius structure" is a technical hypothesis that guarantees finiteness of relevant cohomology spaces. It is almost always satisfied in practice, so it is safe to ignore.
A comment for experts: I do not fix the Frobenius structure as a part of the data of the isocrystal. I only assume that one exists. In fact, here and in several places throughout my research statement, the stated result is likely true more broadly with strictness or non-Liouville hypotheses replacing this assumption.

2.C. **Relating physical and cohomological rigidity.** Let $X = \mathbb{P}^1$. Crew has shown that an irreducible and cohomologically rigid $E \in \textsc{Isoc}^\dagger(U)$ is also physically rigid [Cre17], and his method of proof is (superficially) identical to Katz's proof of the analogous statement for local systems when $k = \mathbb{C}$ [Kat96]. Katz's proof of the converse, however, is transcendental.

Bloch and Esnault give an algebraic proof of the converse in the setting of integrable connections [BE04]. Suppose $k = \mathbb{C}$ and E is an integrable connection on U. The first step in their proof is to consider the deformation functor parameterizing infinitesimal deformations of E with the same "local behavior" along S as E. They identify $H^1(U, j_{!+}\operatorname{End}^0(E))$ as the tangent space of this deformation functor, and then prove that this deformation functor is prorepresentable and smooth when E is irreducible.

Deforming overconvergent isocrystals, however, is a bit subtle. For example, when $U = \mathbb{G}_m$ with coordinate t and dual derivation ∂, one might expect (in analogy with the situation for integrable connections) that deformations of the trivial overconvergent isocrystal on $\mathbb{G}_m$ are those defined by the differential operator $\partial - \alpha/t$ for $\alpha \in \bar{K}$, or perhaps for α in some disk in $\bar{K}$ centered at 0. But it turns out that these differential operators define overconvergent isocrystals precisely if $\alpha \in \mathbb{Z}_p$. In other words, it seems like the space of all deformations of the unit object in $\textsc{Isoc}^\dagger(\mathbb{G}_m)$ is not even an analytic space!

I propose bypassing this difficulty by working in the larger category of all finite differential A-modules, where A is the ring of overconvergent functions on U. Doing this, I construct a deformation functor $\overline{\textsc{Def}}_E^{\sharp,\circ}$ parameterizing infinitesimal deformations of E as a differential A-module with the same "local behavior" along S, and find that it has the desired properties.

Theorem 2.3. *Suppose $X = \mathbb{P}^1$ and $E \in \textsc{Isoc}^\dagger(U)$. There is an explicit natural deformation functor $\overline{\textsc{Def}}_E^{\sharp,\circ}$ whose tangent space is $H^1_p(U, \operatorname{End}^0(E))$.*

(a) If $\operatorname{End}^0(E)$ has a Frobenius structure, then $\overline{\textsc{Def}}_E^{\sharp,\circ}$ has a hull.
(b) If E is also absolutely irreducible, then $\overline{\textsc{Def}}_E^{\sharp,\circ}$ is smooth and prorepresentable.

Thus cohomologically rigid isocrystals are those which have no nontrivial infinitesimal deformations with the same "local behavior" along S. To prove this, I generalize some observations in [BE04] and work with a categorified form of deformation theory. Aside from providing conceptual clarity, this abstraction also enables me to prove that infinitesimal deformations of E can be suitably "algebraized" (cf. [Agr17, 6.7] for a precise statement).

My first step is to categorify results of Yau [Yau05] and prove that the Hochschild cochain complex governs deformations of modules over arbitrary associative algebras over K.

Theorem 2.4. *Let D be an associative K-algebra and E a left D-module. Then the Hochschild complex $\operatorname{Hoch}_K(D, \operatorname{End}_K(E))$, regarded as a differential graded Lie algebra, governs*[4] *the deformation category parameterizing deformations of E.*

[4]One associates a deformation category to any differential graded Lie algebra L, sometimes called the *reduced Deligne groupoid* of L, and L *governs* any deformation category equivalent to its reduced Deligne groupoid.

Then the following provides the link to rigid cohomology.

Theorem 2.5. *Suppose A is a commutative differential K-algebra and E a finite free differential A-module. Let $D = A[\partial]$ be the ring of differential operators over A and regard E as a left D-module. Then there is an explicit quasi-isomorphism of differential graded algebras*

$$\mathrm{dR}(A, \mathrm{End}_A(E)) \xrightarrow{\sim} \mathrm{Hoch}_K(D, \mathrm{End}_K(E)),$$

where $\mathrm{dR}(A, -)$ denotes the de Rham complex of a differential A-module.

Applying these results to the differential K-algebra of overconvergent functions on U, to the Robba rings along S, and then relating the resulting deformation categories, I obtain the deformation functor of theorem 2.3. It is a work in progress to determine if these results can be used to prove that physically rigidity implies cohomological rigidity for absolutely irreducible isocrystals.

2.D. **Rigid rank 2 isocrystals when $X = \mathbb{P}^1$.** I have also developed an understanding of cohomological rigidity for rank 2 absolutely irreducible isocrystals when $X = \mathbb{P}^1$. First, I can characterize the regular ones as follows, using the Christol-Mebkhout index formula.

Proposition 2.6. *Suppose $E \in \mathrm{Isoc}^\dagger(U)$ has a Frobenius structure and that it is regular and absolutely irreducible of rank 2. Then*

$$\dim H^1_p(U, \mathrm{End}^0(E)) = 2 \cdot \#S - 6.$$

Thus E is cohomologically rigid if and only if $\#S = 3$.

My next result is a classification analogous to Yun's classification of rigid ℓ-adic local systems of rank 2 [Yun14, 3.3].

Theorem 2.7. *Suppose $\mathrm{char}(k) > 2$, $X = \mathbb{P}^1$ and $0, \infty \in S$. Suppose further that $E \in \mathrm{Isoc}^\dagger(U)$ has a Frobenius structure and that it is absolutely irreducible and cohomologically rigid of rank 2. Then, possibly after replacing K with a finite extension, one of the following three cases occurs up to equivalence.*[5]

(1) $S = \{0, 1, \infty\}$ and E_s is regular at all three $s \in S$.
(2) $S = \{0, \infty\}$, E_0 is regular and E_∞ has slopes 0 and 1.
(3) $S = \{0, \infty\}$, E_0 is regular and E_∞ is absolutely indecomposable.[6]

In fact, as in the ℓ-adic setting, I believe that all of these possibilities are realized by "hypergeometric" isocrystals. Suppose that k is finite and that $\pi \in K$ satisfies $\pi^{q-1} =$

[5]An overconvergent isocrystal E on U is equivalent to $E \otimes L$ for any $L \in \mathrm{Isoc}^\dagger(U)$ of rank 1. E is also equivalent to $E|_{U'}$ for any dense open subset $U' \subseteq U$. Finally, E is equivalent to $\sigma^* E$ where σ is an automorphism of $\mathbb{G}_m$. This generates an equivalence relation on the set of isomorphism classes in $\mathrm{colim}_{U \subset X} \mathrm{Isoc}^\dagger(U)$, and an object is cohomologically rigid if and only if an equivalent object is cohomologically rigid.
[6]I expect to also be able to prove that E_∞ is pure of slope 1/2 in this third case.

$(-p)^{(q-1)/(p-1)}$, where $q := \#k$ and $p := \mathrm{char}(k)$. For tuples $\underline{\alpha} = (\alpha_1, \ldots, \alpha_n)$ and $\underline{\beta} = (\beta_1, \ldots, \beta_m)$ in K, Katz introduces the generalized hypergeometric differential operators $\mathrm{Hyp}_\pi(\underline{\alpha}; \underline{\beta})$ of degree $\max\{m, n\}$ [Kat90]. These define absolutely irreducible overconvergent isocrystals admitting Frobenius structures [Miy17]. In forthcoming work, I expect to prove the following.

Conjecture 2.8. *If $m \leq n = 2$, the isocrystal defined by $\mathrm{Hyp}_\pi(\underline{\alpha}; \underline{\beta})$ is cohomologically rigid. In the notation of theorem 2.7, it is of type (1) when $m = 2$, type (2) when $m = 1$, and type (3) when $m = 0$.*

2.E. **Rigid regular isocrystals on elliptic curves.** Let us turn now to the situation when X has genus 1. In [Kat96], Katz classifies cohomologically rigid local systems when $k = \mathbb{C}$. I hope to produce the following analogous classification for isocrystals when $\mathrm{char}(k) > 0$.

Conjecture 2.9. *Suppose X has genus 1 and $E \in \mathrm{Isoc}^\dagger(U)$ is absolutely irreducible and regular of rank n, where $\mathrm{char}(k)$ does not divide n.*

(a) If $S = \{s_0\}$, then E is cohomologically rigid if and only if $E \simeq \pi_* L$ for some finite étale cover $\pi : Y \to X$ of degree n and $L \in \mathrm{Isoc}^\dagger(Y \smallsetminus \pi^{-1}(s_0))$ of rank 1.

(b) If $S = \{s_0, \ldots, s_k\}$, then E is cohomologically rigid if and only if $E \simeq L \otimes (E'|_U)$ where $L \in \mathrm{Isoc}^\dagger(U)$ is of rank 1 and $E' \in \mathrm{Isoc}^\dagger(X \smallsetminus \{s_0\})$ is cohomologically rigid.

Presently, I am able to prove the following p-adic analog of one of the results Katz obtains en route to his classification.

Proposition 2.10. *Suppose X is of genus 1 and $E \in \mathrm{Isoc}^\dagger(U)$ is regular and absolutely irreducible. Then E is cohomologically rigid if and only if E_s is "scalar" for every $s \in S$: i.e., there exists an A_s-basis for E_s on which $t\partial$ acts via the scalar matrix*

$$\begin{pmatrix} \lambda_s & & \\ & \ddots & \\ & & \lambda_s \end{pmatrix}$$

for some $\lambda_s \in \mathbb{Z}_p$, where t is a coordinate function on A_s and ∂ the corresponding derivation.

3. Future Directions

3.A. **Middle convolution for isocrystals.** Suppose $X = \mathbb{P}^1$. For both ℓ-adic local systems in [Kat96] and integrable connections in [Ari10], a key part of the middle convolution algorithm is the Fourier transform. Huyghe has developed a theory of the Fourier transform in the setting of arithmetic D-modules as well [Huy04]. This can likely be leveraged to construct a middle convolution algorithm for isocrystals that would explain how to build up rigid isocrystals from rank 1 objects

3.B. **Connections to localization theory.** Suppose $X = \mathbb{P}^1$ again. The localization theory of Huyghe, Patel, Schmidt and Strauch [HPSS15] provides a p-adic analog of Beilinson-Bernstein localization: arithmetic D-modules on a smooth formal lift $\mathfrak{X}$ are equivalent to modules over Emerton's analytic distribution algebra $D^{\mathrm{an}}(\mathbb{G}^\circ)$ with central character 0, where $\mathbb{G}^\circ$ is the "wide open" analytic group over K whose L-points over any completely valued field L extending K are given by

$$\mathbb{G}^\circ(L) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : |a-1|, |b|, |c|, |d-1| < 1 \right\}.$$

The analytic distribution algebra $D^{\mathrm{an}}(\mathbb{G}^\circ)$ is an "overconvergent" analog of the universal enveloping algebra $U(\mathfrak{gl}_2)$.

Banach space representations of $\mathrm{GL}_2(K)$ over K naturally induce modules over $D^{\mathrm{an}}(\mathbb{G}^\circ)$, and Colmez's Langlands correspondence [Col10] relates Banach space representations of $\mathrm{GL}_2(K)$ over K to 2-dimensional Galois representations of $\mathrm{Gal}(\bar{K}/K)$ over K. It would be interesting to work out how rigidity translates through these correspondences.

3.C. **Connections to Abe's Langlands correspondence.** Suppose now that X is arbitrary but that k is finite and $p := \mathrm{char}(k)$. Let F be the function field of X and $\mathbb{A}_F$ the ring of adèles. Yun defines a notion of rigidity for ℓ-adic automorphic representations of $\mathrm{GL}_r(\mathbb{A}_F)$, where $\ell \neq p$, and shows that it corresponds to rigidity of ℓ-adic local systems under the ℓ-adic Langlands correspondence for function fields [Yun14]. In particular, he explains in detail how irreducible rigid ℓ-adic local systems of rank 2 when $X = \mathbb{P}^1$ correspond to rigid automorphic representations of $\mathrm{GL}_2(\mathbb{A}_F)$ under the Langlands correspondence.

Now under Abe's p-adic Langlands correspondence [Abe13], p-adic automorphic representations of $\mathrm{GL}_r(\mathbb{A}_F)$ correspond to rank r isocrystals. It should be possible to work out a notion of rigid p-adic automorphic representations of $\mathrm{GL}_r(\mathbb{A}_F)$ analogous to Yun's ℓ-adic one, and to describe the representations corresponding to the isocrystals described in 2.D.

3.D. **Generalization to G-isocrystals.** If $k = \mathbb{C}$ and G is a reductive group over $\mathbb{C}$, G-local systems on U^{an} are equivalent to exact tensor functors from $\mathrm{Rep}(G)$ into the category of local systems on U^{an}. When $\mathrm{char}(k) > 0$, we pull "the old French trick of turning a theorem into a definition" and define *G-isocrystals* on U to be exact tensor functors $\mathrm{Rep}(G) \to \mathrm{Isoc}^\dagger(U)$ for any split reductive group G over K.

We can then formulate generalized notions of rigidity in this setting: for example, a G-isocrystal $E : \mathrm{Rep}(G) \to \mathrm{Isoc}^\dagger(U)$ is *cohomologically rigid* if $H^1_p(U, \mathrm{Ad}^{\mathrm{der}}(E)) = 0$, where $\mathrm{Ad}^{\mathrm{der}}(E) := E(\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}])$. I can generalize small parts of my foundational results mentioned earlier to this setting: for example, if G is non-abelian, then cohomologically rigid G-isocrystals can only exist when X has genus 0 or 1.

When $k = \mathbb{C}$, Simpson's conjecture predicts that rigid G-local systems are motivic. Beyond Katz's affirmation of this conjecture for $G = \mathrm{GL}_r$, there are mere islands of understanding

in a vast unexplored ocean (for example, Bogner and Reiter establish Simpson's conjecture for $G = \mathrm{Sp}_4$ [BR13]). When $\mathrm{char}(k) > 0$, it would be extremely interesting to understand what can be said about rigid G-isocrystals.

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